RÉPUBLIQUE DU CAMEROUN
Paix-Travail-Patrie
....
MINISTÈRE DE L'ENSEIGNEMENT SUPÉRIEUR
....
COMMISSION NATIONALE D'ORGANISATION DE L'EXAMEN HIGHER NATIONAL DIPLOMA EXAM (HND)
....

REPUBLIC OF CAMEROON
Peace-Work-Fatherland
....
MINISTRY OF HIGHER EDUCATION
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NATIONAL COMMISSION FOR THE ORGANISATION OF HIGHER NATIONAL DIPLOMA EXAM (HND) EXAM
....

National Exam of Higher National Diploma 2023 Session

Speciality/Option: SOFTWARE ENGINEERING (SWE)
Paper : Systems Analysis and Design
Duration : 4 hours
Credit : 7

ANSWR ALL QUESTIONS

SECTION A: OBJECT MODEL (25 marks)

MCQs each correct answer carries one (01) mark

1. Which of the following UML diagrams has a static view?

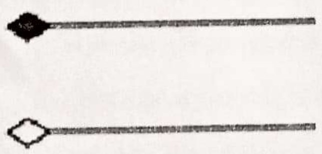
   a) Collaboration

   b) Use case

   c) State chart

   d) Activity

2. What type of core-relationship is represented by the symbol in the figure below?



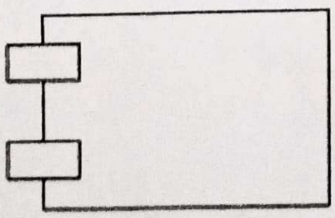
   a) Aggregation

   b) Dependency

   c) Generalization

   d) Association

3. Which core element of UML is being shown in the figure?



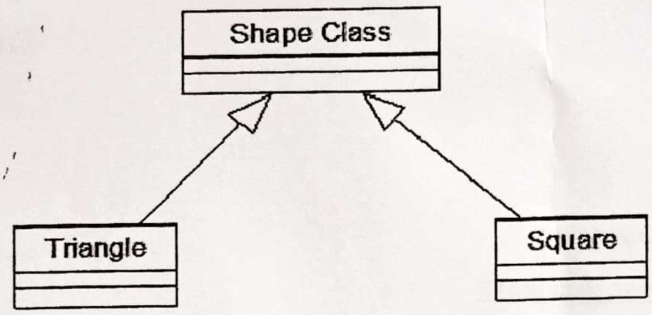
a) Node

b) Interface

c) Class

d) Component

4. What type of relationship is represented by Shape class and Square ?



a) Realization

b) Generalization

c) Aggregation

d) Dependency

5. Which diagram in UML shows a complete or partial view of the structure of a modeled system at a specific time?

a) Sequence Diagram

b) Collaboration Diagram

c) Class Diagram

d) Object Diagram

6. Interaction Diagram is a combined term for

a) Sequence Diagram + Collaboration Diagram

b) Activity Diagram + State Chart Diagram

c) Deployment Diagram + Collaboration Diagram

d) None of the mentioned

7. Which of the following diagram is time oriented?

a) Collaboration

b) Sequence

c) Activity

d) None of the mentioned

8. How many diagrams are here in Unified Modelling Language?

a) six

b) seven

c) eight

d) nine

9. Which of the following is not needed to develop a system design from concept to detailed object-oriented design?

a) Designing system architecture

b) Developing design models

c) Specifying interfaces

d) Developing a debugging system

10. Which of the following is a dynamic model that shows how the system interacts with its environment as it is used?

a) system context model

b) interaction model

c) environmental model

d) both system context and interaction

11. Which of the following is a structural model that demonstrates the other systems in the environment of the system being developed?

a) system context model

b) interaction model

c) environmental model

d) both system context and interaction

12. Which of the following come under system control?

a) Reconfigure

b) Shutdown

c) Powersave

d) All of the mentioned

13. Which model describes the static structure of the system using object classes and their relationships?
   a) Sequence model
   b) Subsystem model
   c) Dynamic model
   d) Structural model
14. Which model shows the flow of object interactions?
   a) Sequence model
   b) Subsystem model
   c) Dynamic model
   d) Both Sequence and Dynamic model
15. Which message is received so that the system moves to the Testing state, then the Transmitting state, before returning to the Running state?
   a) signalStatus()
   b) remoteControl()
   c) reconfigure()
   d) reportStatus()
16. A description of each function presented in the DFD is contained in a ________
   a) data flow
   b) process specification
   c) control specification
   d) data store
17. Which diagram indicates the behaviour of the system as a consequence of external events?
   a) data flow diagram
   b) state transition diagram
   c) control specification diagram
   d) workflow diagram
18. A data model contains
   a) data object
   b) attributes
   c) relationships
   d) all of the mentioned

19. ________ defines the properties of a data object and take on one of the three different characteristics.

a) data object
b) attributes
c) relationships
d) data object and attributes

20. The _________ of a relationship is 0 if there is no explicit need for the relationship to occur or the relationship is optional.

a) modality
b) cardinality
c) entity
d) structured analysis

21. A _________ is a graphical representation that depicts information flow and the transforms that are applied as data moves from input to output.

a) data flow diagram
b) state transition diagram
c) control specification
d) workflow diagram

22. . A data condition occurs whenever a data is passed to an input element followed by a processing element and the result in control output.

a) True
b) False

23. The _________ enables the software engineer to develop models of the information domain and functional domain at the same time

a) data flow diagram
b) state transition diagram
c) control specification
d) activity diagram

24. The _________ contains a state transition diagram that is a sequential specification of behavior.

a) data flow diagram
b) state transition diagram
c) control specification
d) workflow diagram

25. Which of the following is not a construct?

a) sequence

b) condition

c) repetition

d) selection

<u>SECTION B:</u> DATABSE-MYSQL (25marks)

1) What are the three major steps of the database design (data modeling) process? Define each in *one* sentence. **2marks**

2) What types of participation constraints can you have in an E-R model? Define each by *one* sentence. **3marks**

3) What are ACID properties of transactions? Explain each by one sentence. **5marks**

4) **10marks**

You are designing a database for KW Humane Society. The result is the following set of relations where the type of each relations attribute is given following the attribute (e.g., ID: integer):

Animals(ID: integer, Name: string, PrevOwner: string, DateAdmitted: date, Type: string)

Adopter(SIN: integer, Name: string, Address: string, OtherAnimals: integer)

Adoption(AnimalID: integer, SIN: integer, AdoptDate: date, chipNo: integer) where

The primary keys are underlined. Animals stores information about the animals currently at the Humane Society.Each is given an ID, and their names together with the SIN of their previous owners (attribute PrevOwner), and their date of admission is recorded. Type refers to the type of animal (dog, cat, etc). Adopter is the relation that holds information about animal adopters. The attributes are self-descriptive, except OtherAnimals which records the number of other animals that the adopter currently has at home. AnimalID in Adoption refers to the ID of Animals. Similarly, SIN in Adoptionrefers to the SIN of Adopter. Attribute chipNo stores the number on the microchip that is implanted on the animal for tracking. Owner in Animals refers to the SIN of Adopter.

Formulate the following queries in SQL;

a) Retrieve the total number of dogs that were brought to the Humane Society on18 April 2000. **5marks**

b) List the name of the adopter who has adopted every type of animal. **5marks**

c) Define Entity, Entity type, and Entity set. **3marks**

d) What is Data Warehousing? **2marks**

SECTION C: MOBILE OPERATING SYSTEM (25marks)

1) What is the proper way of setting up an Android-powered device for app development? **5marks**
2) What composes a typical Android application project? **5marks**
3) When is the best time to kill a foreground activity? **5marks**
4) Do all mobile phones support the latest Android operating system? **5marks**
5) What is an action in mobile computing OS? **5marks**

SECTION D: DATABASE ADMINISTRATION WITH MYSQL: 25MARKS

1) What are the commands used in DCL? **5marks**
2) What is a system database and what is a user database? **5marks**
3) What is Replication? **5marks**
4) If you are given access to a SQL Server, how do you find if the SQL Instance is a named instance or a default instance? **5marks**
5) What are the different authentication modes in SQL Server and how can you change authentication mode? **5marks**